

Energy Demand in the Central African Republic:

Preliminary results using the CEEDA Tool

Sunil Malla and Govinda R. Timilsina

9 April 2026

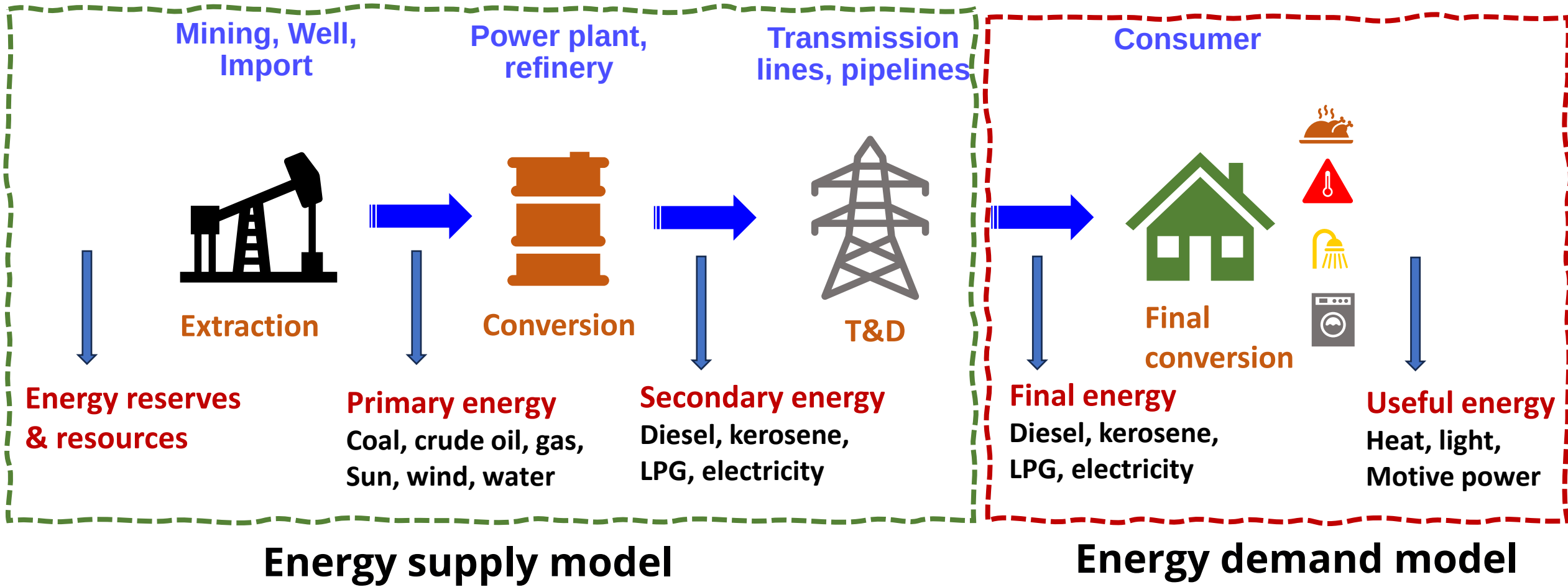
Presentation Outline

1. Energy System and Modeling
2. CEEDA
 - 2.1 Motivation and Key Features
 - 2.2 Framework and Methodology
 - 2.3 Limitation
3. Data, Scenarios, and Preliminary Results
4. Model Demonstration

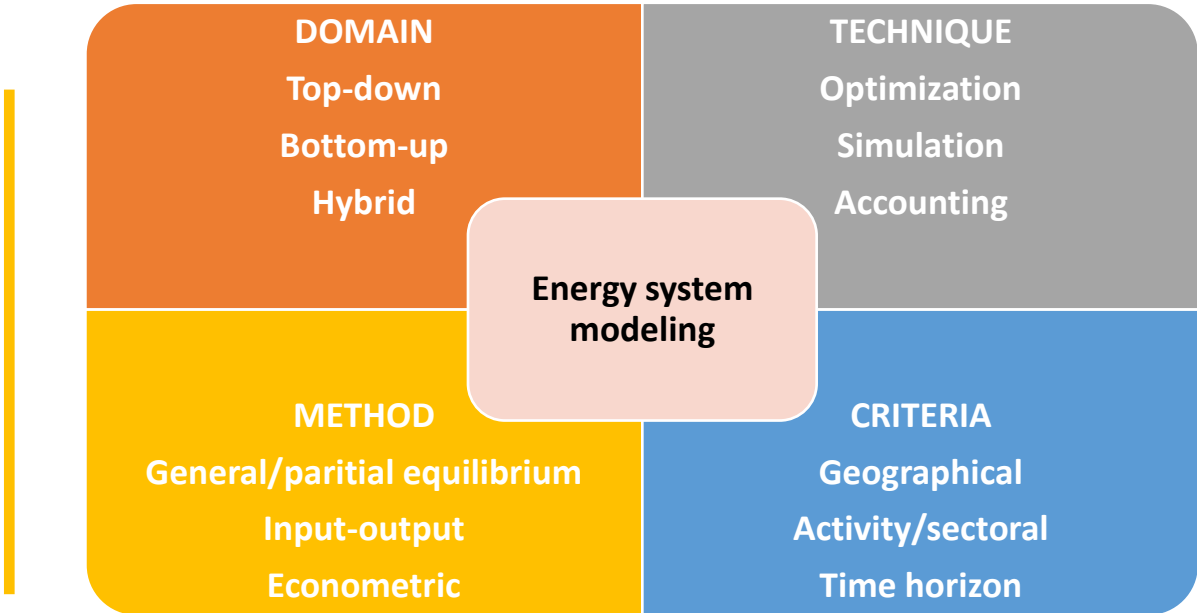
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Energy system modeling and terminology



Energy System Modeling

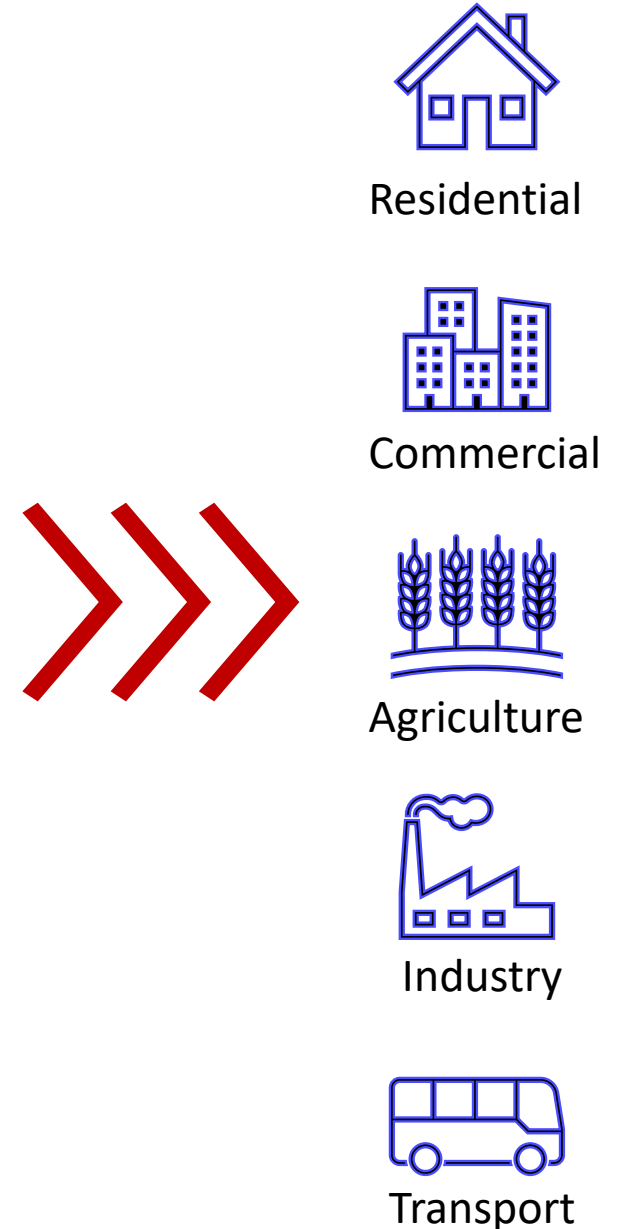
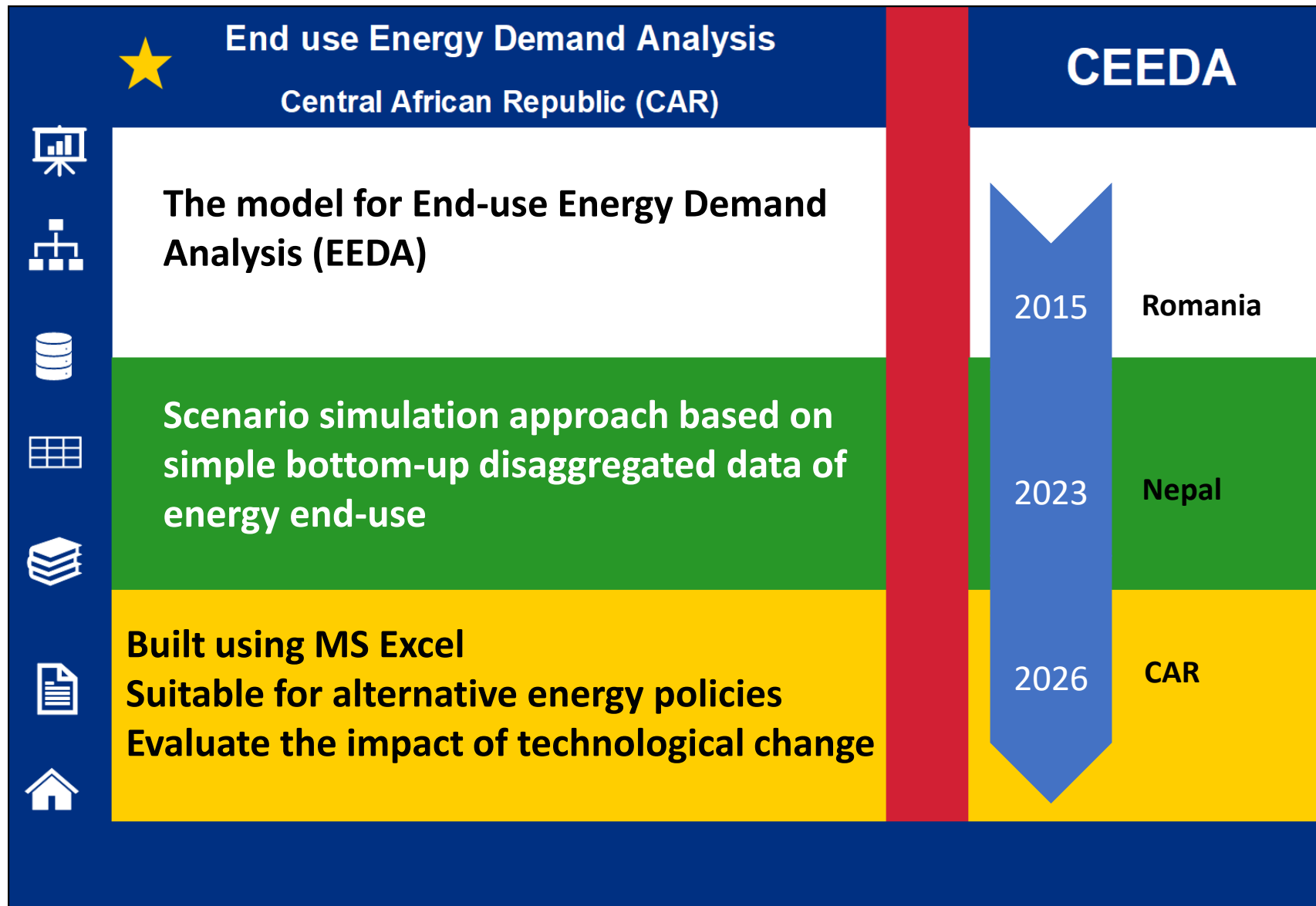


Model	Approach	Method	Horizon	Coverage	Availability	Software	Developer
AIM-Enduse	Bottom-up	S, LP, PE	Up to 2100	National (Global)	Free	GAMS/PATH	NIES (Japan)
EnergyPlan	Bottom-up	S	1 year	National	Free	Stand alone	AU (Denmark)
GEM-E3	Top-down	CGE	2050	Global (38)	NA	GAMS/PATH	EU
LEAP	Hybrid	S, LP	Up to 50 years	National (Global)	Partial free	SA	SEI (Sweden)
MARKAL/TIMES	Bottom-up	LP, PE	User defined	National (Global)	Commercial	GAMS/VEDA	IEA
MESSAGE	Hybrid	LP	Up to 100 years	Global (11)	Upon request	GAMS/ORACLE	IIASA
NEMS	Hybrid	S, O, PE	2050	National (US)	Free	Multiple	EIA
MAED	Bottom-up	A	Up to 40 years	National	Open source	EXCEL	IAEA
POLES	Hybrid	PE/S	Up to 2100	Global (66)	NA	NA	Enerdata
PRIMES	Hybrid	PE	Long-term	National (Europe)	NA	NA	E3MLab (Greece)

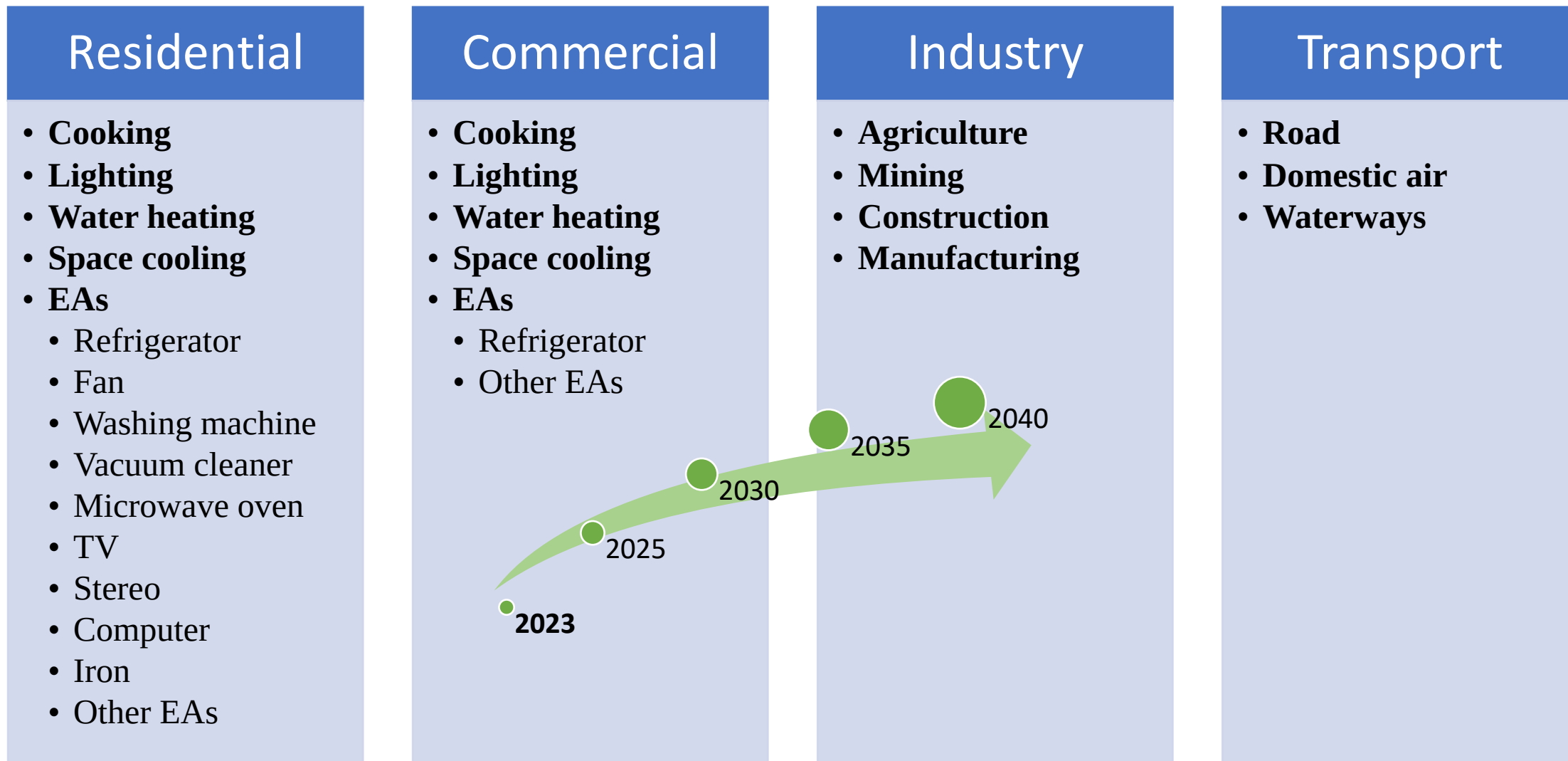
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What is CEEDA?



CEEDA Sectors, Sub-sectors and End-uses

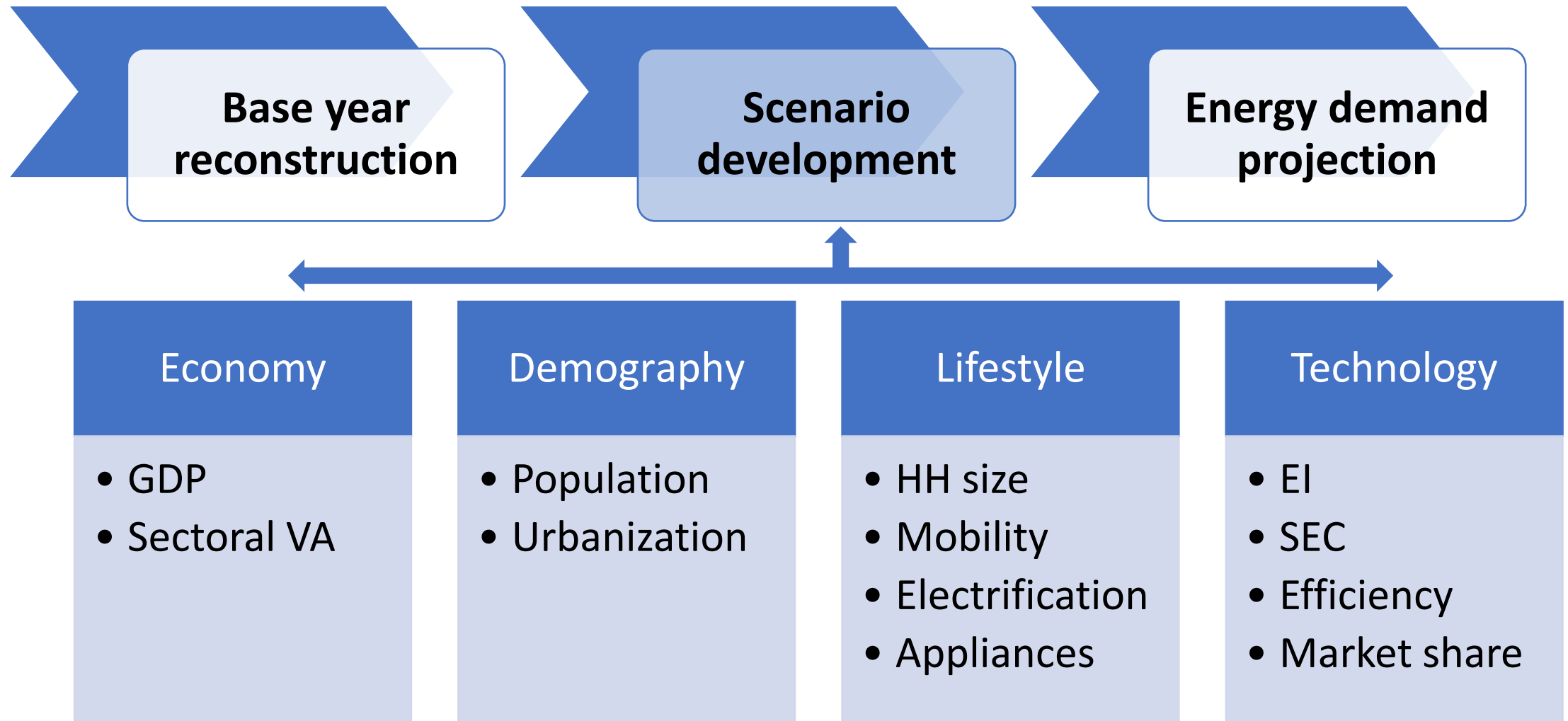


CEEDA Framework (1)

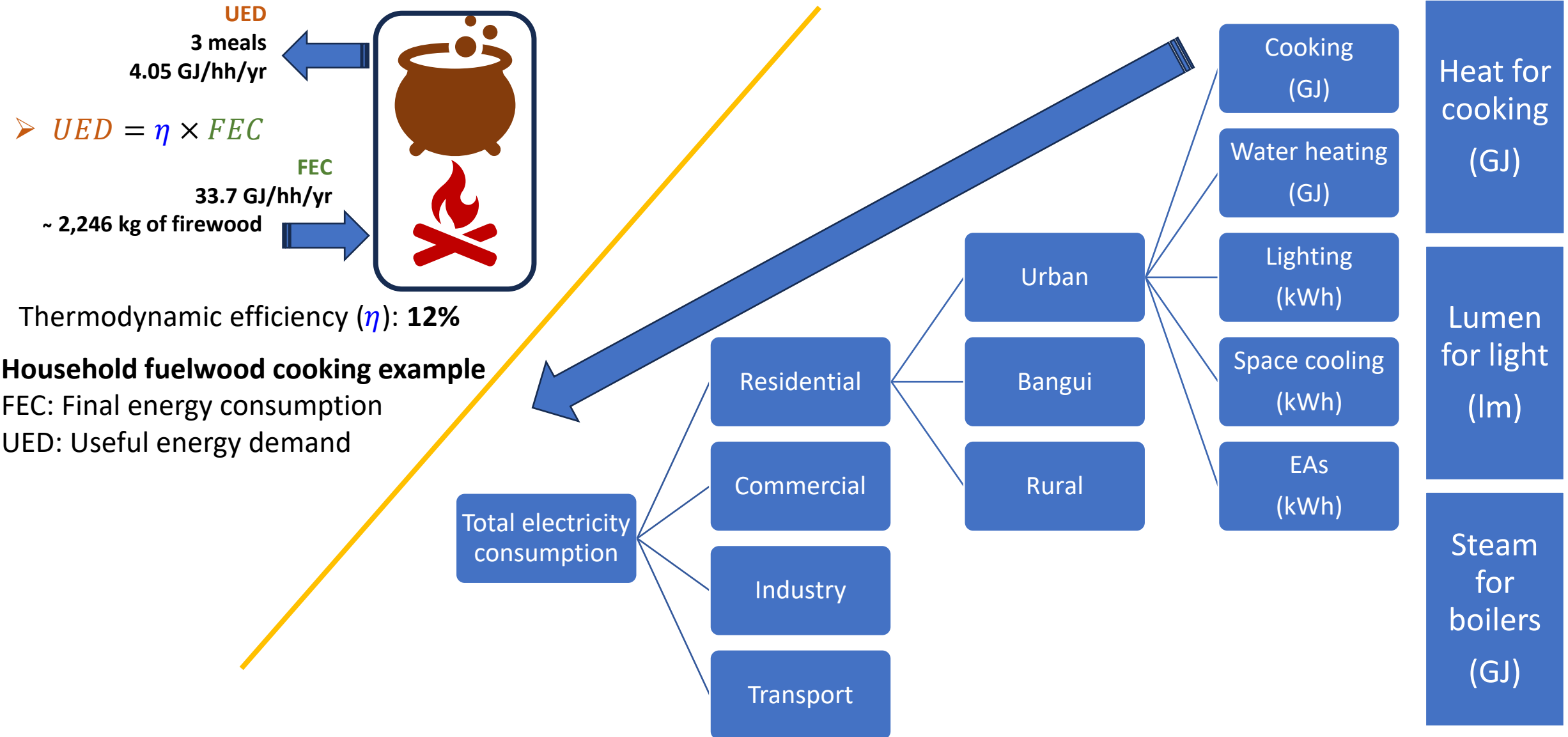


- Reconstruction of the energy consumption in the base year. Total energy consumption by sectors, by end-use, and by energy carriers.
- Establishing scenarios of economic, social, and technological development.
- Analysis of energy demand projections associated with the scenarios.

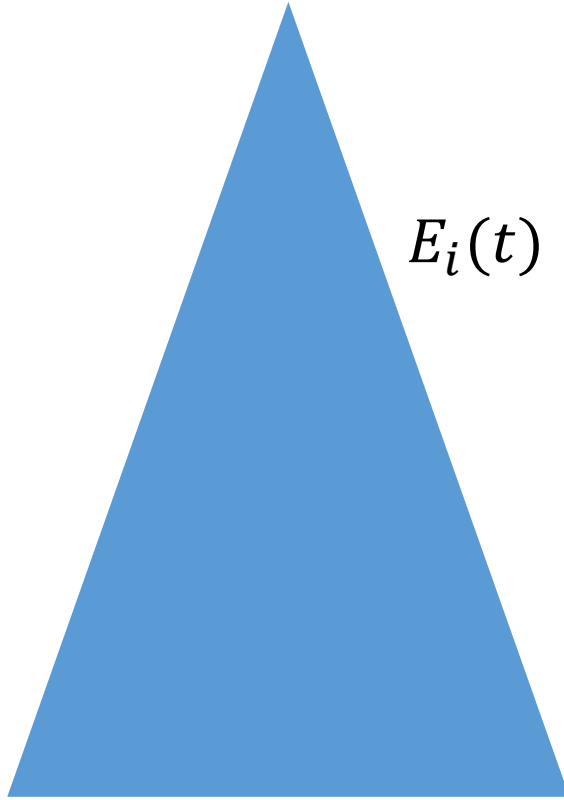
CEEDA Framework (2)



CEEDA Residential Framework (3)



Methodology (E= A X S X I)



$$E_i(t) = A_i(t) \times S_i(t) \times I_i(t)$$

$$E_{i,j}(t) = A_{i,j}(t) \times S_{i,j}(t) \times I_{i,j}(t)$$

A = Activity | S = Structure | I = Energy intensity

i = sector ***j*** = sub-sector or end-use

EEDA Residential model equations

$$E_{r,j}^u(t) = A_r^u(t) \times I_r^u(t) \times \left[\alpha_r(t) \cdot \beta_r(t) \cdot \sum_k^q S_{r,j,k}^u(t) \cdot \frac{1}{\eta_{j,k}^u(t)} \right]$$

where $j \in J \wedge J = \{j_1, j_2, \dots, j_m\}$; $k \in K \wedge K = \{k_1, k_2, \dots, k_q\}$; and $\sum_k^q S_{r,j,k}^u(t) = 1$.

Module I

$$E_r^u(t) = \sum_j^m E_{r,j}^u(t) = A_r^u(t) \times I_r^u(t) \times \left[\alpha_r(t) \cdot \beta_r(t) \cdot \sum_j^m \sum_k^q S_{r,j,k}^u(t) \cdot \frac{1}{\eta_{j,k}^u(t)} \right]$$

such that $\sum_j^m \sum_k^q S_{r,j,k}^u(t) = 1$.

$$E_r^{ea}(t) = \sum_k^q A_{r,k}^{ea}(t) \cdot I_{r,k}^{ea}(t)$$

$$\text{where } I_{r,k}^{ea}(t) = UEC_{r,k}^{ea}(t) = \begin{cases} \alpha_r^k(t) \cdot \beta_r^k(t) \cdot \frac{CAP_{r,k}^{ea}(t)}{EER_{r,k}(t)} & \forall ea = \{ea \in EA \mid ea = ea_1\} \\ \alpha_r^k(t) \cdot \beta_r^k(t) \cdot \frac{CAP_{r,k}^{ea}(t)}{EEI_{r,k}(t)} & \forall ea = \{ea \in EA \mid ea \neq ea_1\} \end{cases}$$

and $EER \in [1, 5]$, $EEI \in [1, 2)$ and ea_1 is space cooling.

	Definition	Unit/Range
Variable		
E	Final energy demand (FED)	kWh
El	Electricity demand	kWh
NEl	Final non-electric energy demand	kWh
I	Intensity	Various
Parameter		
A	Activity	Various
α	Device (technology) operational factor	Fraction
β	Device (technology) utilization factor	Fraction
S	Penetration of fuel/technology type $\forall u$	Fraction
η	Device (technology) efficiency (thermal)	Fraction
CAP	Rated capacity of electric appliance $\forall ea$	kW
EER	Energy efficiency ratio	Fraction
EEI	Energy Efficiency Improvement	Fraction
Index		
u	End-use; $u \in EU$	$EU = \{ck, li, wh, sh, inc\}^\dagger$
ea	Electric appliance; $ea \in EA$	$EA = \{ea_1, ea_2 \dots \dots, ea_l\}^s$
r	Household (individual); $r \in R$	$R = \{r_1, r_2, \dots \dots, r_n\}$
j	Energy type; $j \in J$	$J = \{j_1, j_2, \dots \dots, j_m\}$
k	Technology (device) type; $k \in K$	$K = \{k_1, k_2, \dots \dots, k_q\}$
t	Time interval	2023 (base year); 2025, ..., 2040

$$El_r(t) = \sum_{ea=1}^l E_r^{ea}(t) + \sum_{u=1}^4 E_{r,j_l}^u(t)$$

$\forall j = \{j \in J \mid j = j_1\}$, where j_1 is electricity.

Module II

$$NEl_{r,j}(t) = \sum_{u=1}^p E_{r,j}^u(t) \quad \forall j = \{j \in J \mid j \neq j_1\}$$

For identity, the following relation must hold: $E_r(t) \equiv El_r(t) + NEl_r(t) \equiv \sum_u^4 E_r^u(t) + \sum_{ea}^l E_r^{ea}(t)$

$$E_r(t) = \sum_u^4 E_r^u(t) + \sum_{ea}^l E_r^{ea}(t)$$

$\forall u \in U \wedge U = \{ck, li, wh, sh, in\}$ and $\forall ea \in EA \wedge EA = \{ea_1, ea_2, \dots \dots, ea_l\}$.

EEDA Residential model (ASI)

Definition of household sector intensity, activity, and structure by end-use type.

End-use	Activity (A)	Intensity (I) [†]	Structure (S)
Cooking	Person	$\overline{CK}_r(t)$	
Water heating	Person	$\overline{c} \cdot \overline{m}(t) \cdot [(T_b(t) - T_{a,r}(t)) \cdot \frac{1}{1000}]$ where $\overline{m}(t) = \overline{w}(t) \cdot HHS_r(t)$	$\alpha_r(t) \cdot \beta_r(t) \cdot \sum_j^m \sum_k^q S_{r,j,k}^u(t) \cdot \frac{1}{\eta_{j,k}^u(t)}$
Space heating	Area	$U_r(t) \cdot 24 \cdot HDD_r(t) \cdot \frac{1}{1000}$	
Lighting	Area	$\overline{LI}_r(t) = E_v(t)$	$\alpha_r(t) \cdot \beta_r(t) \cdot \frac{1}{\phi v_{r,j,k}^{li}(t)}$
Electric appliance	Ownership	$\alpha_r(t) \cdot \beta_r(t) \cdot \frac{CAP_{r,k}^{em}(t)}{EEI_{r,k}(t)}$	
Income-generating activity	Household	$\overline{IA}_r(t)$	

Notes: [†] Cooking intensity ($\overline{CK}_r$) is *useful* energy required per person. Water heating intensity is based on specific heat energy formula ($Q = c \cdot m \cdot \Delta T$), where Q is *useful* energy required for water heating, $\overline{c}$ is specific heat value of water, $\overline{m}$ is mass of water, and T_b is base (reference) temperature, and T_a is average annual temperature. Likewise, space heating intensity is based on heat loss rate formula ($\dot{Q} = Area \cdot U \cdot \Delta T$), or $Q/Area = U \cdot 24 \cdot HDD$, where $Q/Area$ is *useful* energy required per unit area for space heating, U is the overall heat-transfer coefficient of building materials, and HDD is heating degree days. Lighting intensity ($\overline{LI}_r$) is illuminance (E_v), measured in lux (lm/area), and dividing this with luminous efficacy (ϕv), measured in lm/W, combined with α and β parameters, gives *final* energy required for lighting. In general, *useful* energy is divided by the conversion efficiency of the end-use device to estimate *final* energy. Intensity for electric appliances, including space cooling, is based on the rated capacity of the electric device (CAP) and energy efficiency improvement factor (EEI), expressed in the *final* electricity required. Income-generating activity intensity ($\overline{IA}_r$) is *final* energy required per household.

Limitation of CEEDA model

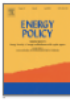
- Based on scenarios. Scenarios are not predictions or forecasts.
- Based on simulation (not optimization)
- Based on bottom-up scenario approach (not econometric)
- Macro-economic factors, such as GDP, are used only as inputs in deriving demand projections without any explicit feedback calculation.
- The price and income are not taken into consideration.

Selected publications



Energy Policy

Volume 38, Issue 4, April 2010, Pages 1979-1990

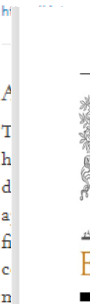


Modelling energy demand of developing countries: Are the specific features adequately captured?

Subhes C. Bhattacharyya ^{a,*}, Govinda R. Timilsina ^b

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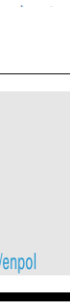


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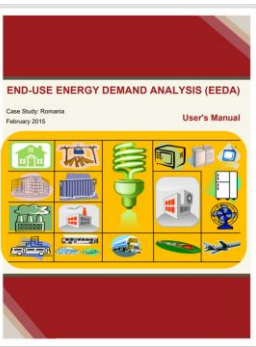
Household cooking in Nepal: An economic analysis

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ARTICLE INFO

ABSTRACT





POLICY RESEARCH WORKING PAPER 11157

Long-Term Energy Demand Forecasting in Romania

Modeling Approach

Sunil Malla
Govinda R. Timilsina



Assessment of long-term sustainable end-use energy demand in Romania*

Sunil Malla ^a and Govinda R. Timilsina ^b

*Flower Mound, TX, USA; ^bDevelopment Research Group, The World Bank, Washington, DC, USA

ABSTRACT
Romania is the 10th largest economy in EU-28 and also one of the fastest growing economies in the region. An end-use energy demand model is developed for Romania to assess energy requirement by sector and by end-use for 2015–2050 period. Industry would surpass residential sector as the largest end-use consumer by 2035. The model is calibrated using 1990–2005 data. The model results show that the total energy demand will increase by 100% by 2050. The model also shows that the energy demand will increase by 100% by 2050. The model also shows that the energy demand will increase by 100% by 2050.

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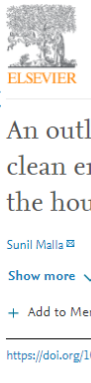


POLICY RESEARCH WORKING PAPER 7697

Long-Term Energy Demand Forecasting in Romania

Modeling Approach

Sunil Malla
Govinda R. Timilsina



Energy

Volume 238, Part B, 1 January 2022, 121810

Economics of Household Cook Electricity in Nepal

Sunil Malla 

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<https://doi.org/10.1016/j.energy.2021.121810>

- ### Highlights
- Household end-use energy demand by province and regions through 2051 is analyzed.
 - Final energy demand reduction is significant under alternative scenarios wrt baseline.
 - Cooking remains major household end-use activities up to 2031 across the scenarios.
 - Electricity is key, but biomass remains significant in future energy-mix under SDG7.
 - Priority provinces/end-uses are identified where clean energy services are lacking.



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Assessment of mobility and its impact on energy use and air pollution in Nepal

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ABSTRACT
Understanding of existing motorized and non-motorized transport services is a key challenge for land-locked Nepal in addressing sustainable transport system development. This paper assesses mobility and estimates current travel demand, energy use and emissions from different transport modes for Kathmandu (capital city) and for rest of the country. Road transport dominates all transport modes in Nepal. In per-capita terms, country's motorized road passenger travel (1461 km) is amongst the lowest in the world. Private vehicles (mainly motor cycle) in Kathmandu and public vehicles (mainly bus) in rest of the country, dominate road passenger travel. Trucks dominate in freight transport services. More than half of country's total commercial energy is consumed by transport sector. Kathmandu alone consumes country's road energy use, based on ever, emissions of local air pollutant e for deteriorating air quality in the xt, transport sector is responsible for



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Patterns and its environmental energy access and poverty in Nepal

and, Nepal

are analyzed based on LEAP framework. s and biomass for cooking dominates country's energy-mix. energy access and poverty across the regions. viating energy poverty in Nepal.

A C T
ly 87% of Nepal's total final energy is consumed by households. This paper analyzes the household energy use and associated air pollutant emissions in Nepal based on LEAP or thirteen analytical regions and three end-uses. Four scenarios involving different growth cio economic and energy system development through the year 2040 are considered. The that household energy use is heterogeneous across the regions and biomass for cooking ie country's energy-mix. Households' CO₂ emissions are less significant but their local indoor issions will continue to rise in the future. To help strengthen government's commitment to ainable energy for all initiative, this study devises an energy development index (EDI) to ry's energy access and poverty across the regions. The results reveal that the current level of access and energy poverty in the country is below the basic human needs and this situation : by little in next 30 years. The paper argues that to improve these situations require more and innovative plans and policies from the government. The paper suggests that greater ll be needed in reducing dependence of biomass for cooking, promoting domestic alternative ces, scaling up biomass improved cookstoves programs and developing periodic regional database.

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Data and Driving Variables in the CEEDA model

Data

- Data categorized into **three** main groups
 - Energy data
 - Socio-economic and demographic data
 - Technical data
- Data **sources**:
 - UN Population and Energy Statistics
 - CAR's Household Survey (EHCVM 2021)
 - World Bank's WDI
 - AEP; UN ECA; IMF
 - Other supplementary reports and studies

Key Driving variables:

- Residential
 - Population, HH size, urbanization
 - Clean cooking fuel share
 - Floor area
 - Electric appliances ownership
- Commercial
 - Share of total VA and growth rate
 - No. of establishments
- Industry/Agriculture
 - Share of total VA and growth rate
- Transport
 - GDP growth rate
 - Penetration of electric road vehicle

Scenario storylines

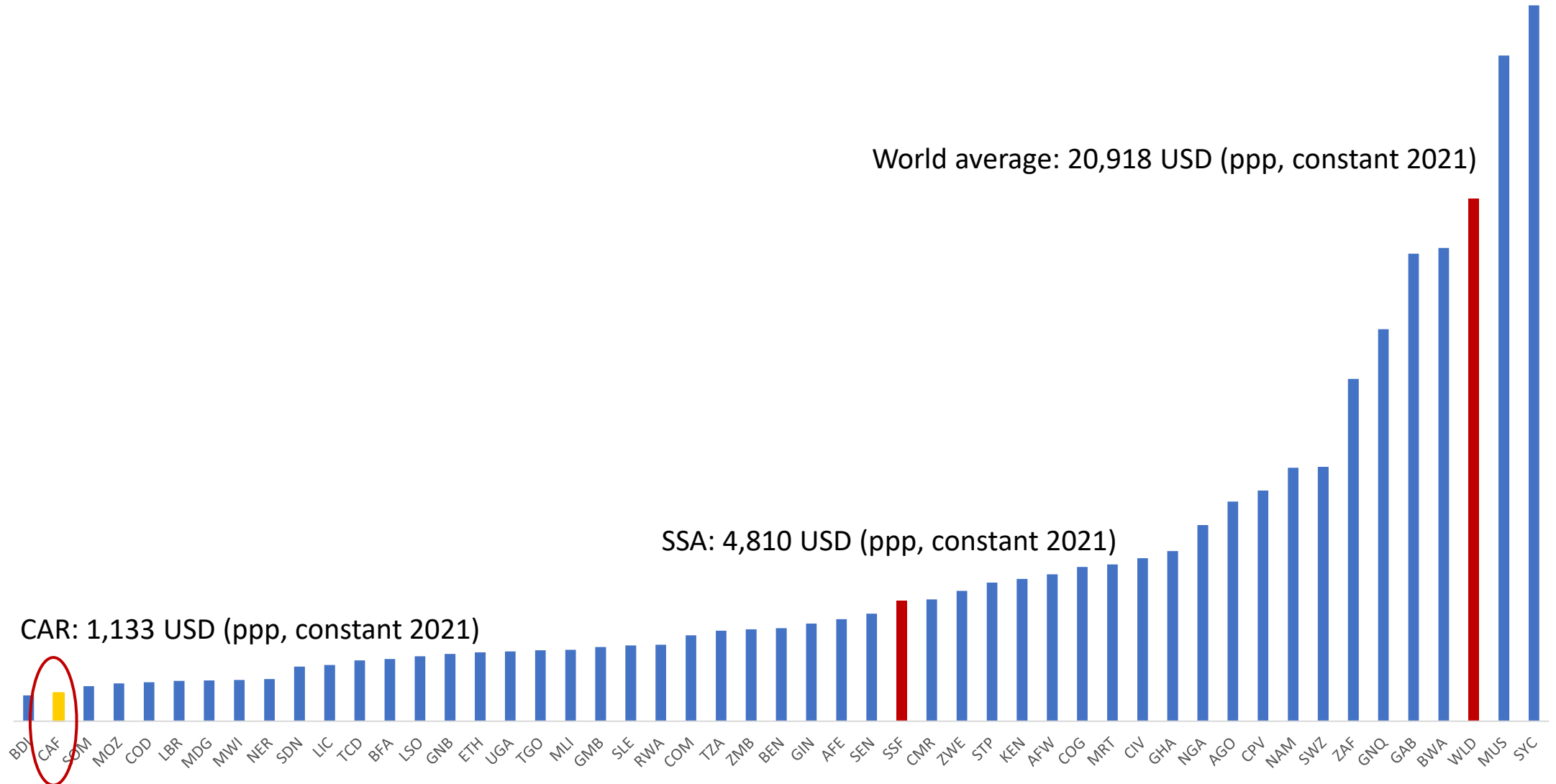
Business-as-usual

- It reflects the path of future energy demand given a **continuation of current and incremental trends**.
- **Medium variant** population growth rate 2.95% (8.5 million in 2040) and urban population growth rate of 3.9% (4.3 million in 2040).
- Clean cooking share [**<1%** (2023) to **23%** (2040)]
- Access to electricity [**16%** (2023) to **33%** (2040)]
- Average annual growth rate
 - Services: 4.6%
 - Industry: 4.1%
 - Agriculture: 1.4%
 - GDP: 3.3%

Optimistic

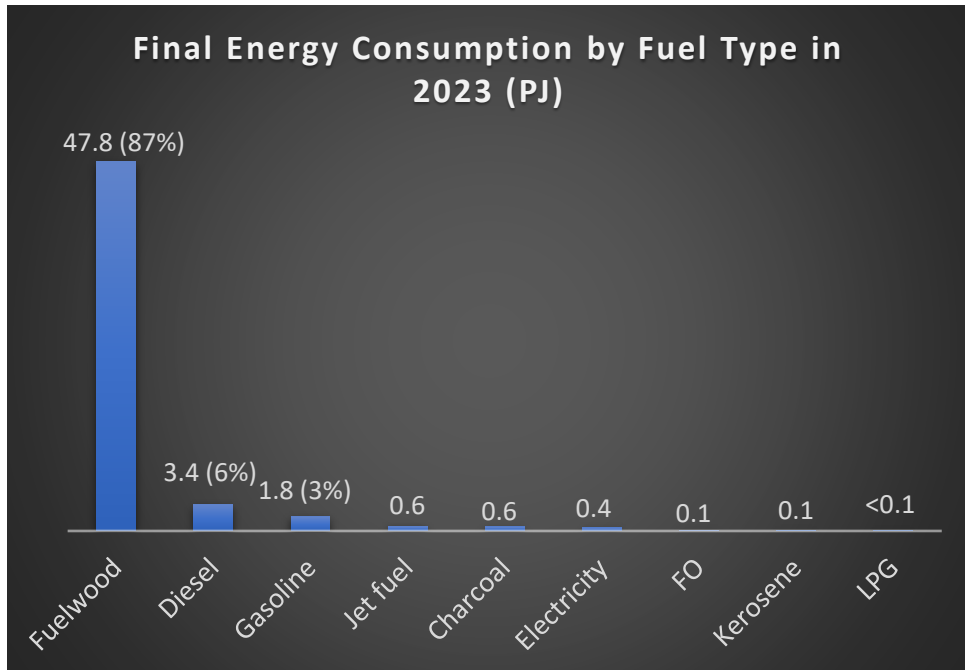
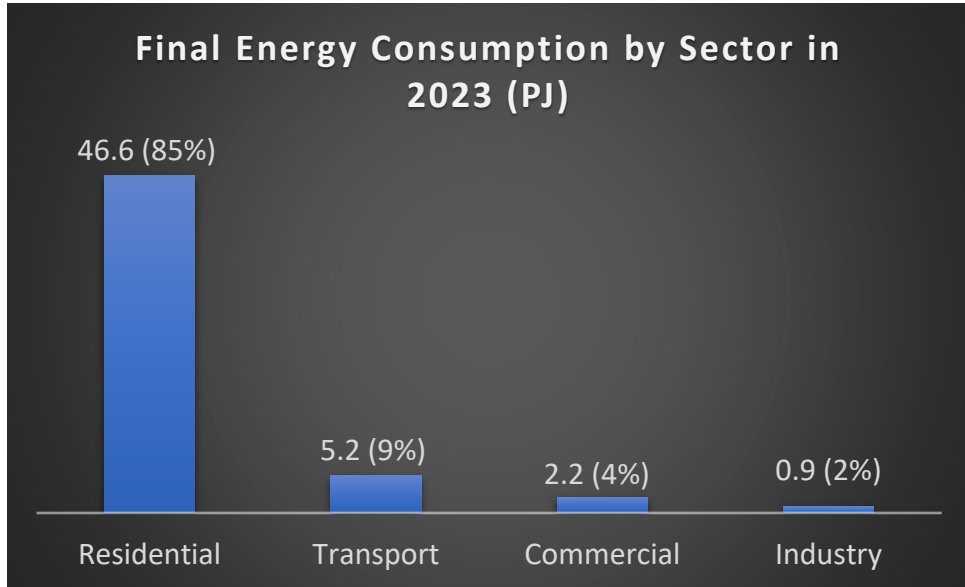
- Envisioned by drawing parallels with the **current trajectory of Cameroon**.
- **Shift** toward **cleaner energy** systems.
- **High variant** population growth rate 3.2% (8.9 million in 2040) and urban population growth rate of 4.2% (4.7 million in 2040).
- Clean cooking share (**31%** in 2040)
- Access to electricity (**65%** in 2040)
- Average annual growth rate
 - Services: 5.7%
 - Industry: 4.2%
 - Agriculture: 3.5%
 - GDP: 4.1%

Per Capita GDP across the African Countries



Data source: WDI, World Bank

Current Energy Consumption Situation in the CAR (2023)

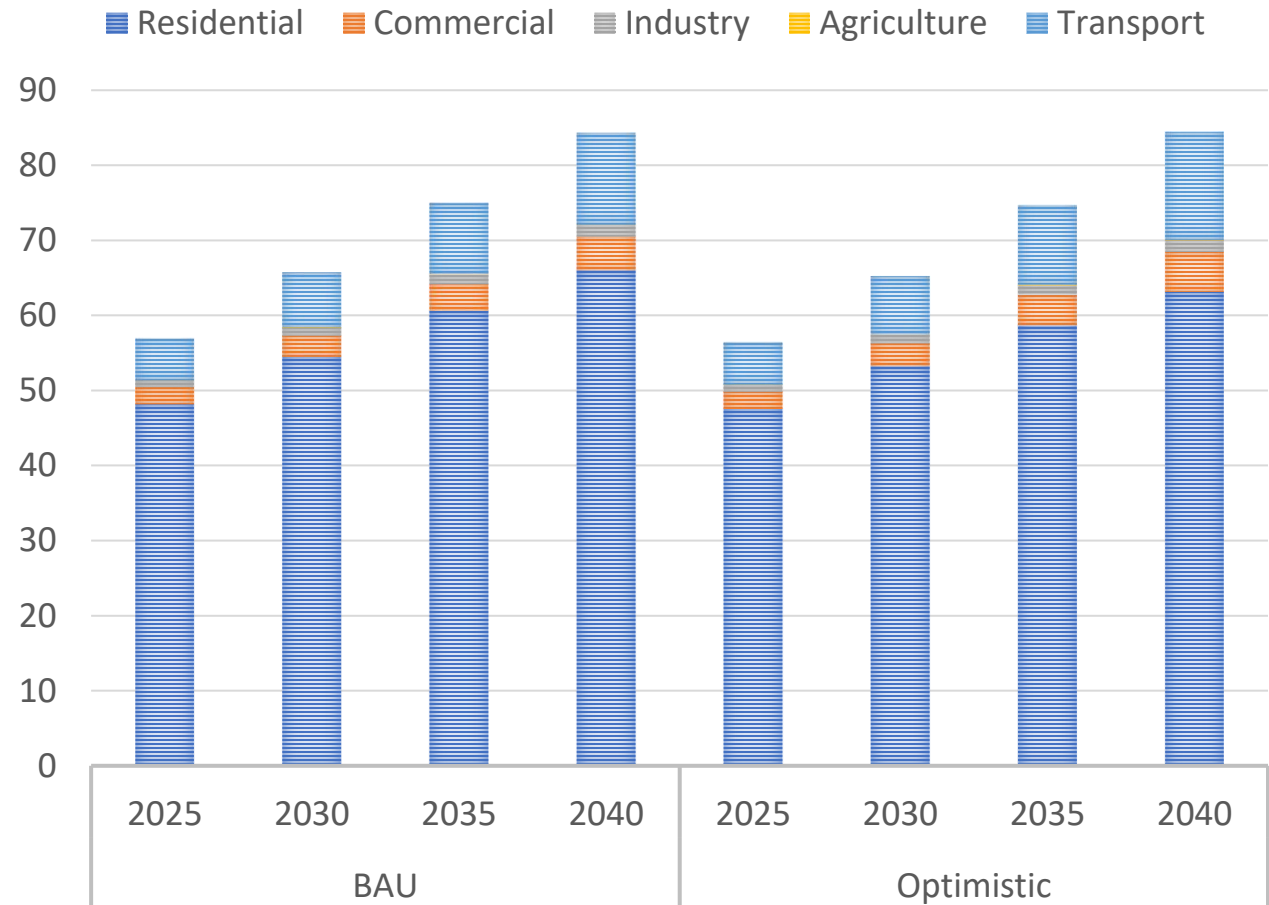


- Extremely **low modern energy access** and a **high dependence** on traditional unprocessed biomass.
- About **16%** of the population has access to grid electricity.
- Electricity consumption per capita is about **24 kWh** (Cameroon: **284 kWh** and Kenya: **190 kWh**)
- The **residential** sector **dominates** final energy consumption.
- This is largely driven by **fuelwood** used for **cooking**.
- Access to **modern cooking fuel solutions** is virtually nonexistent. (CAR: <1%; Cameroon: 31%; Kenya: 32%)
- Current household **ownership of electric appliances** (e.g., refrigerators and air conditioners) is **very low** (< 1%).
- Heavy reliance on traditional biomass likely contributes to **deforestation, indoor air pollution, and adverse health outcomes**.

Energy Demand by Sector

- **Residential** sector remains the **dominant** throughout the period (78% in BAU to 75% in the Optimistic scenario).
- The **transport** sector experiences the most significant increase. More than double from the base year to 2040 under both scenarios.
- The **commercial** sector exhibits steady growth, and the **industry** sector rises progressively.
- Relatively **lower** overall energy demand under the **optimistic** scenario compared to **BAU** is mainly due to a gradual diversification **away from heavy residential reliance on traditional biomass** and increased **contributions of modern fuels** from transport and commercial activities.

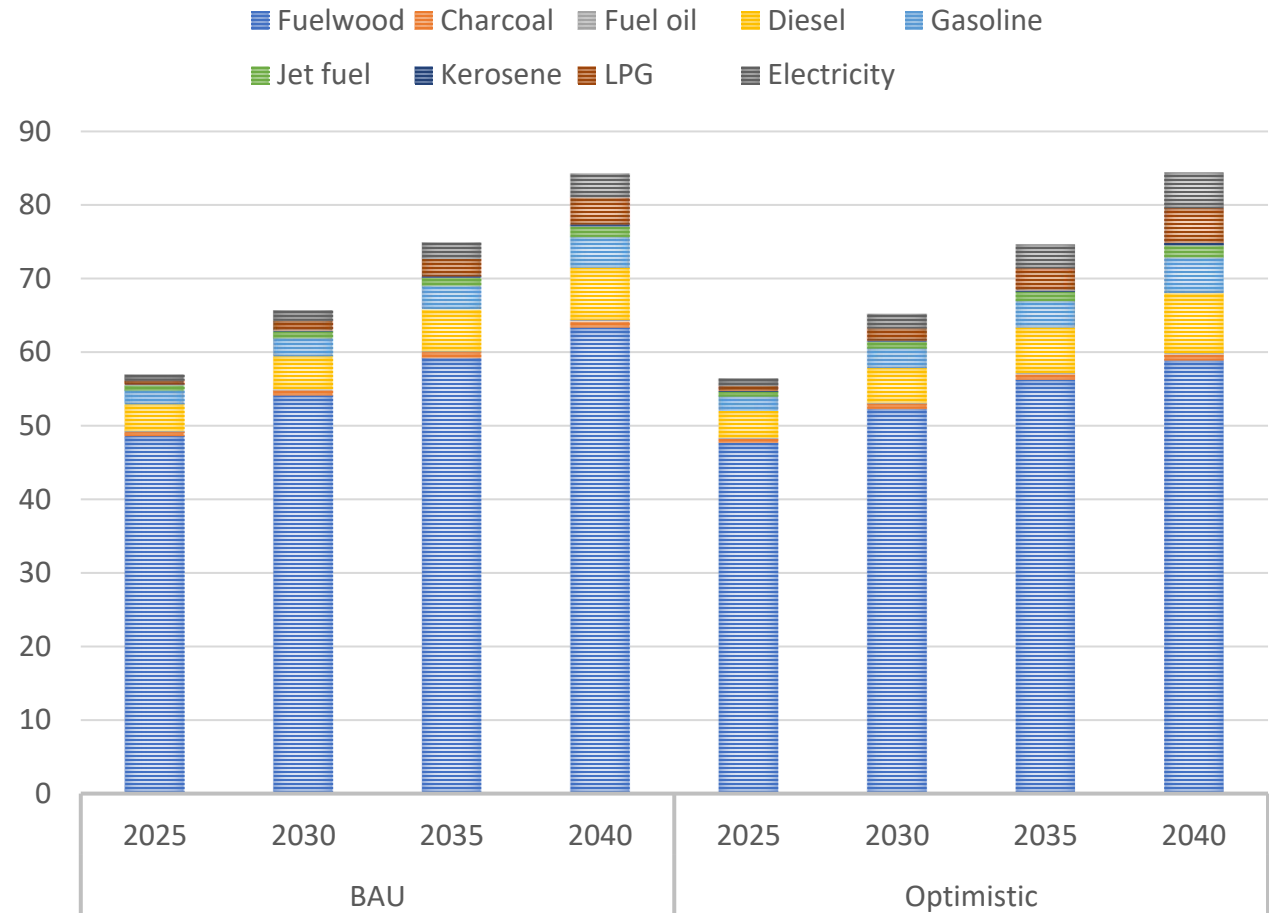
CAR TOTAL ENERGY DEMAND BY SECTOR (PJ)



Energy Demand by Fuel Type

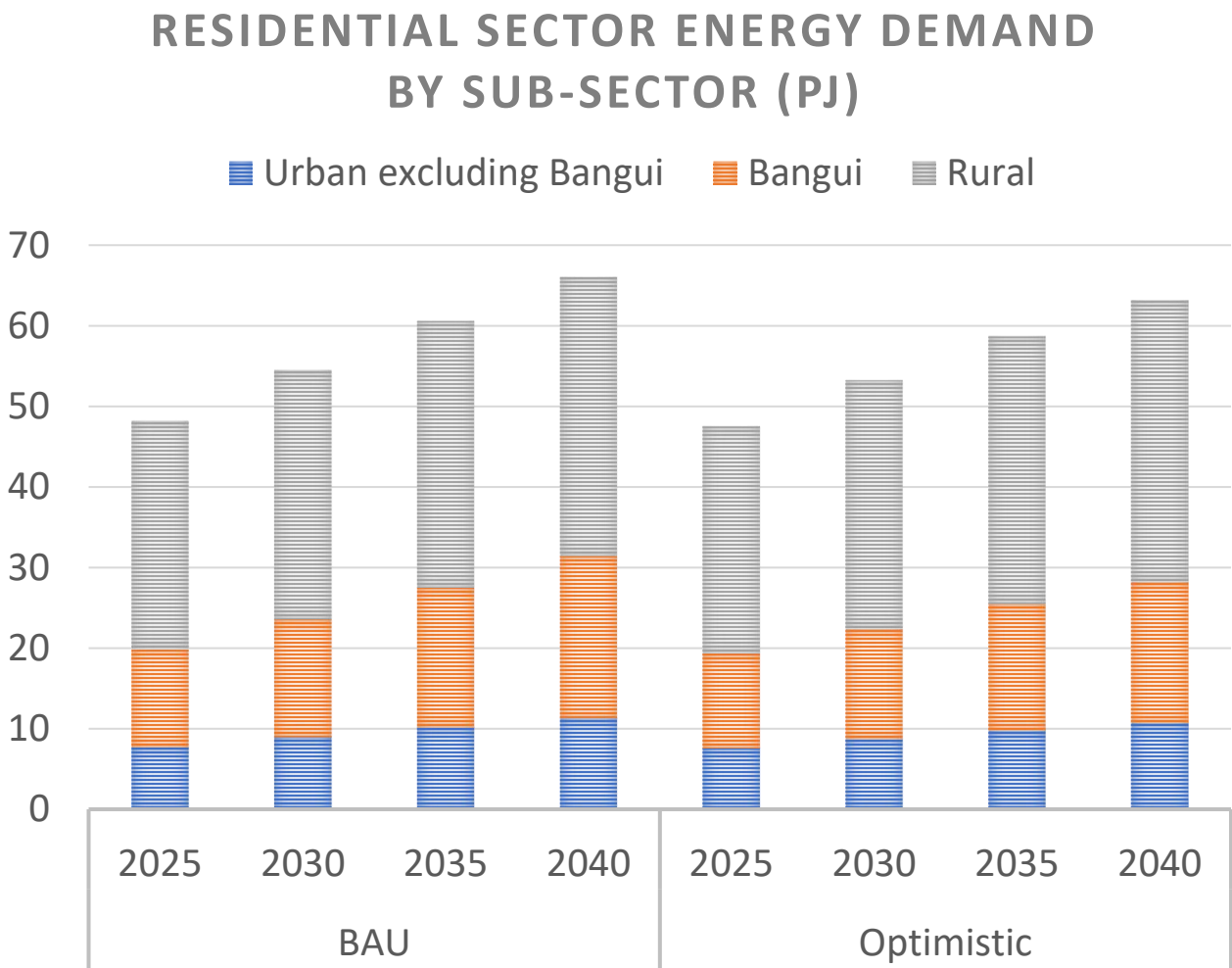
- **Heavy reliance on traditional fuelwood**, mainly used for cooking, but its share declines steadily over time (**75%** in BAU to **70%** in the Optimistic scenario compared to **87%** in 2023).
- **Oil products** show strong growth between 2023 and 2040 under both scenarios, driven largely by expanding **transport demand** and **LPG** in the residential sector.
- Demand for electricity increases significantly. About **seven-fold** under BAU and **eleven-fold** under the Optimistic scenario between 2023 and 2040.

CAR TOTAL ENERGY DEMAND BY FUEL TYPE (PJ)



Residential sub-sector energy demand

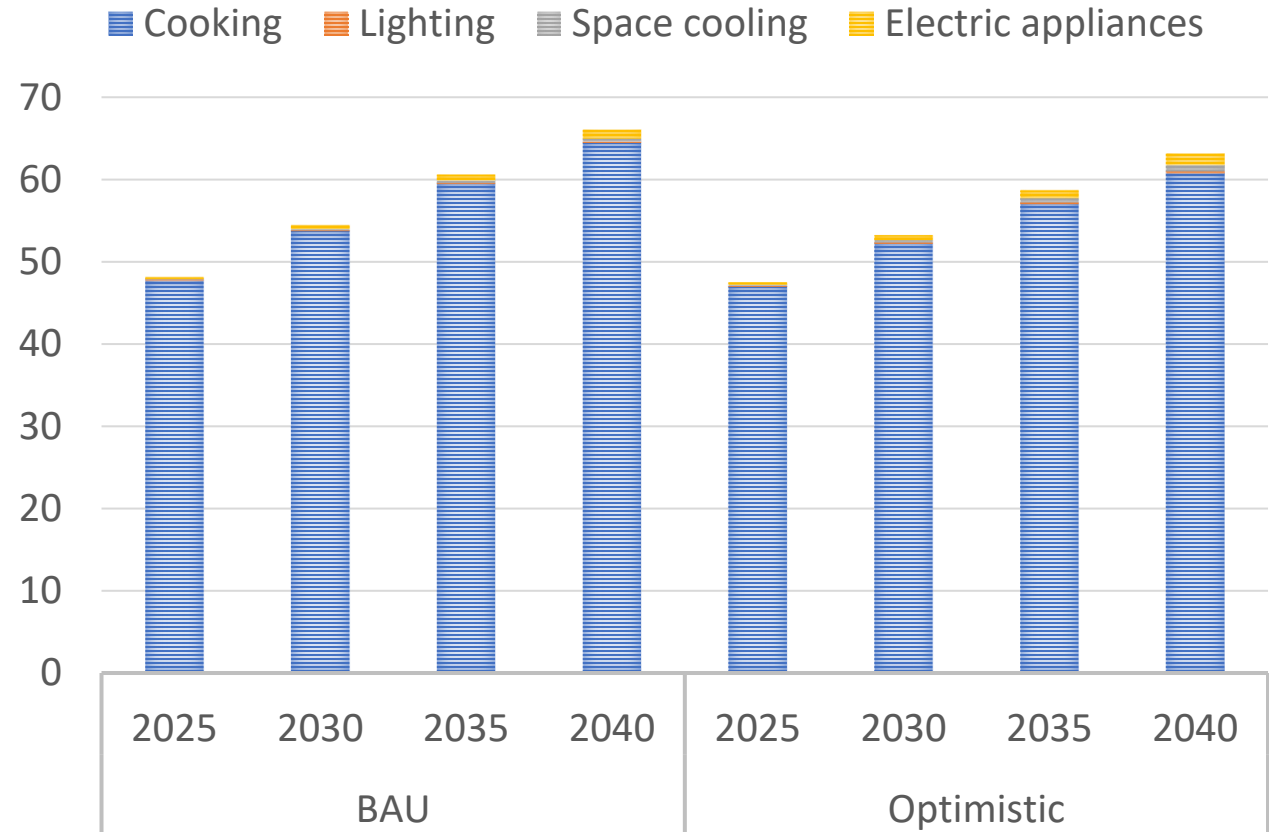
- **Rural** regions consume most of the residential sector's total energy. Ranging from **52%** in BAU to **55%** in the Optimistic scenario, compared to **60%** in 2023.
- This is followed by Bangui and the other rural regions.



Residential end-use energy demand

- Cooking contributes most to the total residential energy demand. Around **97%** under both scenarios.
- There is a substantial increase in the energy demand by **electric appliances** from 0.1 PJ to 1.1 PJ (BAU) and 1.4 PJ (Optimistic) in 2040.

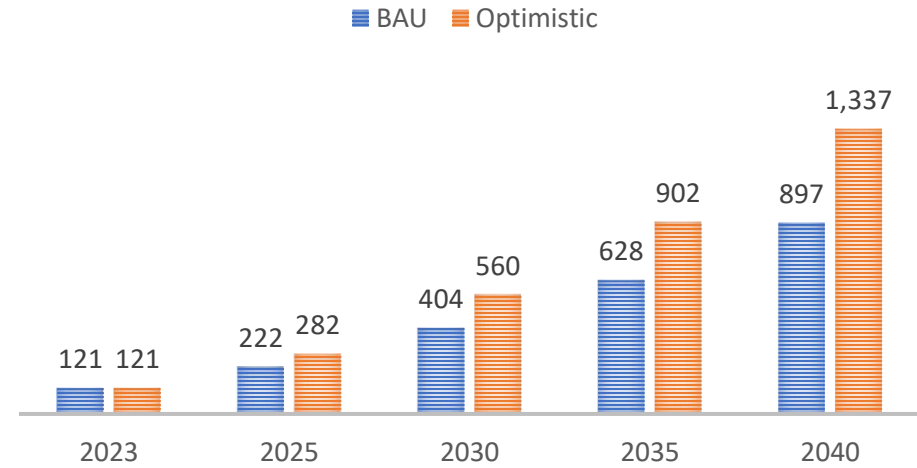
RESIDENTIAL SECTOR ENERGY DEMAND
BY END-USE (PJ)



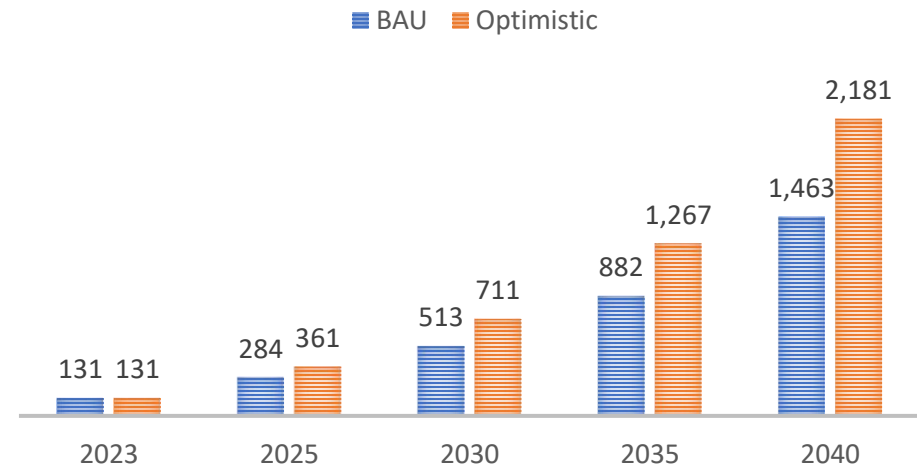
Electricity demand and Installed capacity

- **Increase in total electricity demand**
 - 2023: **121 GWh**
 - BAU: **897 GWh**
 - Optimistic: **1,337 GWh**
- **Increase in total installed capacity**
 - 2023: **131 MW**
 - BAU: **1,463 MW**
 - Optimistic: **2,181 MW**
- **Investment Costs (2040)**
 - BAU: **5.4 billion USD** (2022 price)
 - Optimistic: **8.1 billion USD** (2022 price)

TOTAL GRID ELECTRICITY DEMAND (GWH)



TOTAL GRID INSTALLED CAPACITY (MW)



Initial Assessments

- Currently in the CAR, extremely **low modern energy access** and a **high dependence** on traditional unprocessed biomass.
- The **residential** sector dominates final energy consumption, largely driven by **fuelwood** used for cooking.
- Heavy reliance on traditional biomass likely contributes to deforestation, indoor air pollution, and adverse health outcomes.
- Although small in magnitude, **demand for electricity increases significantly**. About **seven-fold** under BAU and **eleven-fold** under the Optimistic scenario between 2023 and 2040.
- Electricity Generation Investment Costs **5.4** billion USD to **8.1** billion USD.

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End use Energy Demand Model (EEDA)

Central African Republic (CAR)

This tool is to look at the end-use energy demand analysis and forecast in CAR

Color scheme for cells

Sheets	Description
Main Menu	Current sheet
Classification	This sheet contains the sector/end-use of a sector
Data	This sheet contains the input information and variables
Result	This sheet calculates and evaluates the end-use energy analysis
Dashboard	Output
Report	This sheet contains graphs and tables for the report
Reference	This sheet contains information on reference for the input dataset used

Contact



Thank you.

